

**RAJIV GANDHI COLLEGE OF ENGINEERING AND  
TECHNOLOGY**

**COMPUTER SCIENCE AND ENGINEERING**

# ***ASSIGNMENT - 5***

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## Department

## Vision

## And

## Mission

### Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering

### Mission

- To impart quality education in Computer Science and Engineering by means of learning techniques and Value-added courses.
- To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interactions.
- To build leadership skills, ethical values and teamwork among the students.
- To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.



## PART - A

1. What is design pattern? What are the steps of design pattern?

A design pattern is a reusable solution to a common problem that occurs in software design, providing a proven development paradigm to help developers create more maintainable, flexible and scalable software systems.

### Types of design patterns

- Abstract pattern
- Structural pattern
- Behavioural pattern.

2. What are the steps involved to select a design pattern?

- Identify the problem
- Analyze the requirements.
- Research design patterns.
- Evaluate pattern applicability.
- Consider trade-offs.
- Choose a design pattern.

3. When to use patterns and state the use of design patterns?

The design patterns should be used when they help solve common design problems efficiently and improve the quality of the system.

### The use of design patterns

1. Read the pattern once through for an overview.
2. Go back and study the structure, participants and collaborations sections.

3. Look at the sample code section to see a concrete example of the pattern in code.
4. Choose names for pattern participants.
5. Define the classes.
6. Define application specific names for operations in the pattern.
7. Implement the operations to carry out the responsibilities.

4. What is catalog of design pattern?

A catalog of design pattern is a structured collection of common, reusable solutions to recurring design problems in object oriented software design.

→ GOF catalog

→ GRASP pattern

→ Enterprise Integration patterns.

5. List any six pattern template.

- Pattern Name.
- Intent
- Motivation
- Applicability
- Structure.
- Participants

6. What is the use of builder pattern?

The builder pattern is used to construct complex objects step by step, separating the construction process from the final representation.



→ It allows creating different representation of an object using the same construction process.

7. Differentiate pattern and frame work?

### Pattern

A reusable solution to a specific design problem providing a general outline or template for solving a particular issue.

### Framework

A semi complete structure that provides a foundation for building application including predefined components interface and relationships.

8. List out the importance of chain of responsibility?

1. Decouples sender and receiver

The patterns allows multiple objects to handle a request without the sender needing to know which object will handle it.

2. Improves flexibility and scalability.

It adds or changes handlers easily without modifying existing code, supporting open/closed principle.

9) What is mean by low coupling?

Low coupling is refers to a design principle where classes or modules have minimal dependencies on each other, making the system more modular, flexible and maintainable.

→ This reduces the impact of changes in one module on other modules.

10. How to apply the GRASP patterns?

1. Identify responsibilities

Determine what each object should be based on system behaviour.

2. Assign responsibility

Use GRASP principle (like Information Expert, Creator, Controller) to assign responsibilities to appropriate classes for better design and maintainability.

### PART-B

11. Write a short note on the following patterns.

- a) ~~Abstract~~ pattern
- b) Structural pattern
- c) Behavioural pattern.

a) Abstract pattern

→ Abstract design patterns abstract the instantiation process.

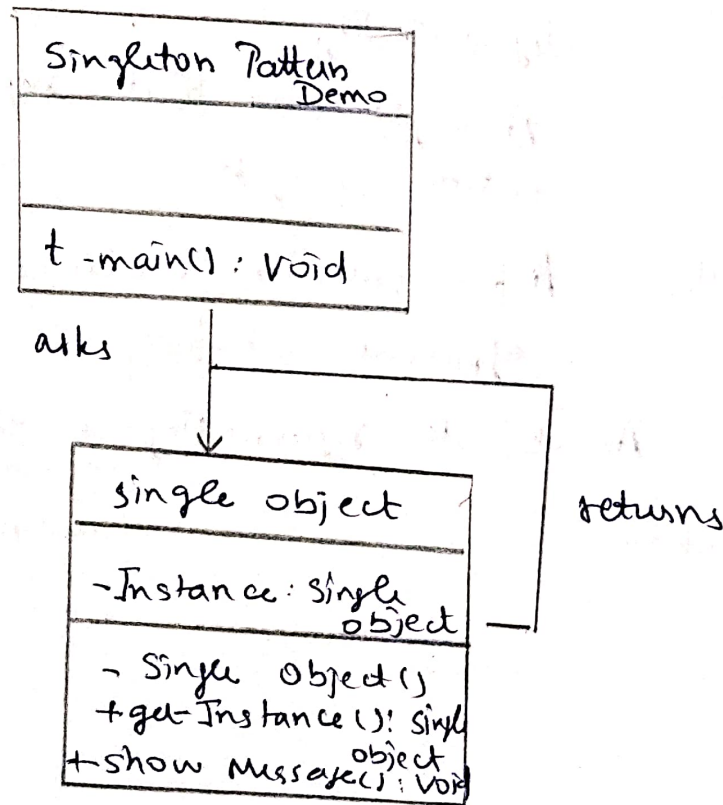
They help make a system independent of how its objects are created composed and represented.

→ A class creational pattern uses inheritance to vary the class that's instantiated, whereas an object creational pattern will delegate instantiation to another object.

Different types of creational pattern are as follows:-

→ Abstract Factory pattern - Creates an instance of several families of classes.

- Builder pattern - Separates object construction from its representation.
- Factory method pattern test:  
Creates an instance of several desired classes.
- Prototype pattern - A fully initialized instance to be copied or cloned.
- Singleton pattern - A class of which only a single instance can exist.



## b) Structural pattern

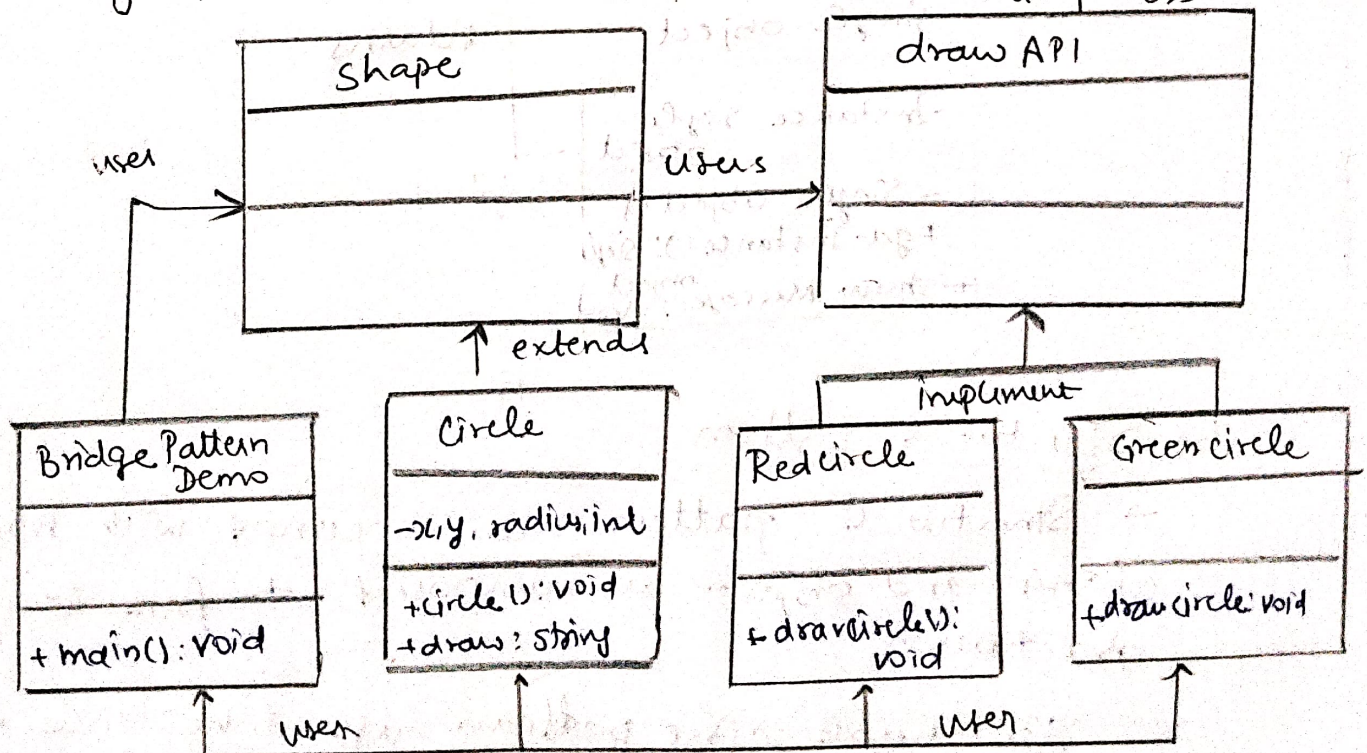
- Structural patterns are concerned with how classes and objects are composed to form larger structures.
- Structural class patterns use inheritance to compose interfaces or implementations.
- As a simple example, consider how multiple



inheritance mix two or more classes into one.  
The result is a class that combines the properties of last parent class.

Different types of structural patterns are as follows.

- Adapter pattern - Match interfaces of different class.
- Bridge pattern - Separates an object's interface from its implementations.
- Composite pattern - A tree structure of simple and composite objects.
- Decorate patterns - Add responsibilities to objects dynamically.
- Facade patterns - A single class that represents an entire subsystems.
- Flyweight pattern - A fine grained instance used for efficient sharing.
- Proxy pattern - An object representing another object <interface>



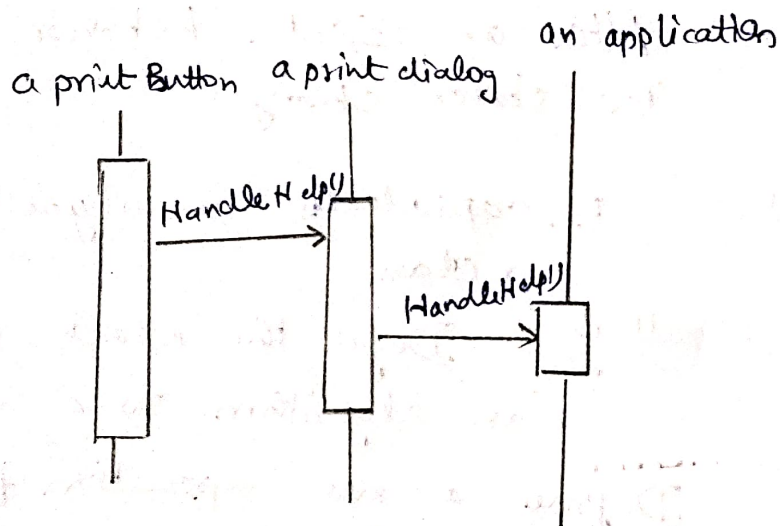


### c) Behavioural pattern

The first object in the chain receives the request and either handles it or forwards it to the next candidate on the chain. The object that made the request has no explicit knowledge of who will handle it - we say the request has an implicit receiver.

Let's assume the user clicks for help on a button widget marked 'print'. The button is contained in an instance of 'print dialog', which knows the application object it belongs to.

The following interaction diagram illustrates how the help request gets forwarded along the chain:



Behavioural patterns are those patterns that are most specifically concerned with communication between objects.

#### → Chain of Resp. pattern

A way of passing a request between a chain of objects.

#### → Command pattern - Encapsulate a command request as an object.

12. Describe the categories of design pattern. State the role and pattern while developing system design?

- Design patterns are reusable solution to commonly occurring problems in software design.
- They are not code, but templates or 'blue prints' that can be applied to solve design issues in various contexts.
- Making code more flexible, reusable and maintainable.
- Communicating design efficiently between developers.
- Solving common architecture problems.

### Categories of Design patterns

There are three main categories they are as follows:

- Creational pattern.
- Structural pattern.
- Behavioral pattern.

Let us discuss in details about such categories

#### Creational pattern:

→ Deal with object creation mechanism, trying to create objects in a manner suitable to the situation.

→ Some of the types of creational pattern

- \* Singleton
- \* Factory Method
- \* Abstract Factory
- \* Builder
- \* Prototype



## Structural patterns

→ Concerned with how classes and objects are composed to form layer structures.

→ Some of the types of structural patterns.

- \* Adapter
- \* Bridge
- \* Composite
- \* Decoder
- \* Facade
- \* Proxy
- \* Fly weight

## Behavioural patterns

→ Focus on communication between objects how they interact, and distribute responsibility.

→ Some types of behaviour pattern are as follows.

- \* Observer
- \* Strategy
- \* Command
- \* State
- \* Template, method
- \* Mediator
- \* Iterator
- \* Chain of responsibility

Role of Design patterns in system Design:

- Improve code reusability.
- Enhance system scalability and Flexibility
- Promotes maintainability.
- Enables better communication